ACDM flavored fried chicken version of Galaxy Dynamics

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Abstract

This is a research experiment. The author does not believe the ΛCDM recipes, halos or particle readings below. It works out how a ΛCDM group could reverse-engineer the law, so that the route is on record first. Nothing here is a result of the author's framework or feeds it. "Fried chicken" is a nickname and carries no physics. This paper tunes the knobs of ΛCDM galaxy modelling, supernova feedback first, until halos reproduce the law $g_N = g^2/(g + a_0)$, with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}} = 9.36 \times 10^{-11}\,\mathrm{m\,s^{-2}}$ and $\kappa = \frac{1}{2}$ fitted, on the measured baryons of 171 SPARC galaxies. It is easy to get close. Twelve feedback knobs, as one global recipe and with no knob set by a galaxy's own curve, land the fitted acceleration within 0.3 % of a_0 and sit 0.0415 dex from the law, 0.0445 on galaxies held out of the tuning. Against the observed rotation curves that recipe scores 0.1953 dex, and the law 0.1951. Adding two concentration knobs gives 0.0275 dex and 0.1933. Six knobs per galaxy, inside 2σ priors, reach 0.0152 dex and pass the 0.02 dex target frozen in advance, though at least half the galaxies then sit at the low bound on concentration. The law's tie to dark energy did not come from the knobs. It took five terms written in by hand, at extreme values, and κ still narrowly missed. A halo can also be written to equal the law exactly, including the tie, by defining it from the formula. That passes every target by construction, and it needs a coupling to dark energy that ΛCDM does not contain. On KiDS-1000 lensing, the law, the recipes and that halo all fail. Version 1.2 tries for a complete ΛCDM recipe, using the law as its basis, against a frozen scoreboard. One global recipe with 26 knobs and hand terms is not complete. Every line can be passed somewhere only by a patchwork of about 1066 parameters, which is not one theory. It also fits X-COP and KiDS better than the plain law does. These recipes were tuned to the law, so a match means freedom, not explanation. This is not evidence for the law or against ΛCDM, and every knob is disclosed.

1 What is being imitated

The target is the law's curve on each galaxy's own baryons, not the observed rotation curve. The law is not new in form: the scale goes back to Milgrom [13, 14], and this interpolating form has appeared before [15]. Only the coefficient looks new, and it is fitted. The law carries its own failures, including IC 1613 and X-COP, listed in PAPER67 (DOI 10.5281/zenodo.23883741). A perfect imitation inherits them.

SPARC [16] supplies the baryons, with $\Upsilon_{\rm disk} = 0.5$, $\Upsilon_{\rm bul} = 0.7$ and no per-galaxy adjustment, giving 3373 points. Four of the 175 galaxies have too few usable points. Against the observed curves on these points, the law's rms is 0.1951 dex.

2 The recipe: twelve supernova knobs

Galaxy simulations are run with subgrid feedback models that carry many tunable parameters. We can't run hydrodynamic simulations here, so R11 is an analytic stand-in, and the mapping from knob to halo is ours, not any simulation's parameter file.

- **Stellar mass outcome, 4 knobs.** The stellar-to-halo relation [21]: normalisation, $\log M_1$, β, γ.
- **Feedback energy, 4 knobs.** $e = \epsilon\,(M_*/0.01M_{200})^{a_1}\,(\Sigma_b/100)^{a_2}\,(f_{\rm gas}/0.5)^{a_4}$, and the core exponent is $n = \tanh e$ [20].
- **Core size, 2 knobs.** $\eta = \eta_0(M_*/10^9)^{a_3}$, core radius $\eta\,R_{1/2}$.
- **Contraction or expansion, 2 knobs.** $f = f_0 + f_1 \log_{10}(M_b/10^{9.5})$ [19].

Halos are Einasto with $\alpha = 0.16$ and $c(M)$ from [22]. The tuned values are log norm -0.319, $\log M_1$ shift $+0.705$, β shift -0.884, γ shift -0.600, $\log\epsilon = -0.639$, $a_1 = -0.760$, $a_2 = +0.091$, $a_4 = -0.097$, $\log\eta_0 = +1.000$, $a_3 = +0.466$, $f_0 = +1.000$, $f_1 = -0.335$. Three sit at the bounds of the ranges we chose: γ, η_0 and f_0. Halo masses stay near abundance matching, with $\log M_{200}$ shifted by -0.23, $+0.11$, $+0.30$ at the 16th, 50th and 84th percentiles.

The recipe gives $a = 9.331 \times 10^{-11}$. Tuned on half the galaxies, it scores 0.0405 dex there and 0.0445 on the other half, so it transfers. With $c(M)$ norm and slope also free (R11b, 14 knobs), it reaches 0.0275, and 0.0326 on unseen galaxies, with M_{200} up by a median of 0.39 dex and both c knobs at their bounds.

3 Targets, frozen before any number

E1: pooled rms to the law at most 0.02 dex. E2: 95th percentile of per-galaxy rms at most 0.04 dex. E3: no half-dex $g_{\rm bar}$ bin off by more than 0.02 dex. E4: $d\ln a/d\ln\rho_{\rm DE} = 0.50 \pm 0.05$ at fixed ω_m, ω_b, A_s. E5: $a(z)/a(0)$ inside the law's band at $z = 1$, 1.5 and 2.33. Most recipes here were added after first numbers, at the owner's request, and are marked Amendments A1 to A3 in `FROZEN_RECIPE.md`. The targets never moved. The knob ranges and priors are our judgement.

4 The menu

The level comes cheaply. Four knobs bring the fitted acceleration from 15.76 to 9.12×10^{-11}, within 3 % of a_0, with the concentration knob at its bound. Getting the right number from tuned halos is easy, so the number alone proves little.

The shape does not. Halos set by mass alone can't follow galaxies of the same mass and different surface density. Coupling c and M_{200} to M_b/R_d^2 helps, to 0.0408 dex, with the abundance-matching normalisation at its bound. It still misses all three targets. Measured against the observed curves, R7 has rms 0.1964 dex, R11 0.1953, and R11b 0.1933, against the law's 0.1951. None was tuned to those curves. With these fixed mass-to-light ratios, this statistic barely tells them apart, so a good fit to rotation curves alone cannot pick the law over a tuned halo.

The thermal halo misses. An isothermal halo in the baryon potential looked like the natural imitator [31]. With two global laws it fails at 0.2245 dex, from either starting point, and with 342 free parameters it still reaches only 0.0609. Its per-galaxy fits do land near $\sigma^4 = GM_b a_0/4$, median ratio 1.017, with wide scatter (0.568 to 3.174).

Recipe	Parameters	rms	p95	worst bin	a (10^{-11})
R1 NFW, abundance matching, no tuning	0	0.1419	0.2877	0.1613	15.76
R1 tuned, with contraction	4	0.0715	0.1486	0.0441	9.12
R2 tuned, with feedback core	6	0.0721	0.1357	0.0283	8.65
R3 thermal halo, two global laws	2	0.2245	0.3187	0.2261	6.06
R4 thermal halo, per galaxy	342	0.0609	0.1119	0.0260	8.97
R7 surface-density coupling	9	0.0408	0.0777	0.0413	9.21
R11 twelve feedback knobs	12	0.0415	0.1006	0.0221	9.33
R11b feedback plus c norm, slope	14	0.0275	0.0624	0.0196	9.31
R10 six knobs as global functions	42	0.0213	0.0441	0.0216	9.30
R9 per galaxy, 1σ priors	1026	0.0376	0.0711	0.0164	9.63
R9 per galaxy, 2σ priors	1026	**0.0152**	**0.0283**	**0.0040**	9.43
R8p per galaxy, referee priors	684	0.0396	0.0665	0.0148	9.72
R8 per galaxy, wide	684	**0.0056**	**0.0130**	**0.0031**	9.36
R6 one empirical function	1	**0.0075**	**0.0118**	**0.0180**	9.37
R5 exact shell masses	3373	**0.0000**	**0.0000**	**0.0000**	9.36
R12 halo defined from the formula	1	**0.0000**	**0.0000**	**0.0000**	9.36

Table 1: Distance to the law in dex. Bold passes E1 to E3. Thresholds are 0.02, 0.04 and 0.02.

5 The exact dish, and its bill

R9 is the closest match on ΛCDM parts inside stated priors: per galaxy M_{200}, c, Einasto α, contraction fraction, core exponent and core radius. With 2σ priors (0.22 dex in c, 0.6 dex in M_{200}) it passes E1 to E3 at 0.0152 dex. The bill: median $\log c$ shift -0.22, at the bound, and median α 0.1, also at the bound. Scatter can't push half the galaxies to the same edge, so this is really a population shift in c. With 1σ priors it fails, at 0.0376. NFW [18], Blumenthal contraction [19], a coreNFW core [20]. Each galaxy gets its own M_{200}, c, contraction fraction and core exponent. R8 passes E1 to E3. The bill, relative to abundance matching [21] and $c(M)$ [22]: $\log M_{200}$ shifts by median $+0.29$ and $+1.2$ at the 84th percentile, and $\log c$ by median -0.42, reaching -0.81 at the 16th. Contraction is near full (median 0.9). Under priors of 0.11 dex in c and 0.3 dex in M_{200}, R8p is 0.0396 and fails E1 and E2.

R10 writes those six knobs as quadratic functions of M_b, surface density and gas fraction: 42 coefficients, one recipe. It stops just short at 0.0213 dex, and 0.0270 on unseen galaxies, but with median $\log M_{200}$ shift $+0.46$, $\log c$ shift -0.74, and 0 of 171 galaxies inside 1σ of both relations.

R6 is one line, the published empirical form [17], with $g_\dagger = 9.585 \times 10^{-11}$, or 1.024 a_0. It passes E1 to E3 with one number, but writes the law's content as a halo response with no process behind it.

R5 is the law renamed: $M_{\rm DM}(r) = r^2(g_{\rm law} - g_{\rm bar})/G$. We checked whether it could exist at all, as a spherical halo with baryons smoothed by 0.1 dex. Density is positive in 171 of 171, the isotropic Jeans dispersion is positive in 171 of 171, and the Eddington distribution function is non-negative to 1 % of its peak in 171 of 171. The same Eddington code returns $+0.0000$ for Hernquist and -0.2555 for a cored tracer in a point-mass potential, so it can fail. Stability is therefore no argument against the imitation.

Its origin is the weak point. Undoing Blumenthal contraction gives a monotonic initial profile in 158 of 171 galaxies. An NFW fit to that initial profile has median rms 0.057 dex and 0.261 at the 84th percentile, with median c at 0.32 of the $c(M)$ value. Against abundance matching, the implied $\log M_{200}$ has median $+0.16$, but -0.76 at $\log M_b$ 11 to 12.

6 The dark energy tie

This is the part the knobs don't supply. On R10 with no hand term, the slope with $\rho_{\rm DE}$ is +0.283, κ runs 0.459, 0.614, 0.735 across $(\Omega_\Lambda, H_0) = (0.70, 67.4)$, $(0.685, 73)$, $(0.65, 73)$, and $a(z)/a(0)$ is 1.58, 2.12, 3.44 at $z = 1$, 1.5, 2.33. Five terms written in by hand change that. $c \propto \rho_{\rm DE}^{3.00} H_0^{-1.61}$ and $M_{200} \propto H_0^{-12}$ give slope +0.496, which passes, and κ of 0.485, 0.506, 0.525, a narrow miss of the 5 % target. Two of those exponents sit at the bounds we allowed, and no process produces them. $c \propto (1+z)^{-0.26-0.89z}$ gives 0.93, 0.91, 0.92, inside the band, with baryons held fixed in redshift, which no real galaxy does. So the tie can be imitated by hand, almost, and then it is the law's physics written into the halos. R5 and R6 pass E4 and E5 only if a_0 is written into them directly. The thermal halo R3, with the same scale written in, gives +0.810 and fails.

7 The halo exactly like the formula

R12 is the reverse-engineered dish: one function for every galaxy, added after the owner asked for it and frozen as Amendment A6.

$$M_{\rm DM}(r) = \frac{r^2}{G}\left[g_{\rm law}(g_{\rm bar}; a) - g_{\rm bar}\right], \qquad a = \kappa c\sqrt{G\rho_{\rm DE}}, \quad \kappa = \tfrac{1}{2} \text{ by hand.} \tag{1}$$

In ΛCDM's language this is a halo of collisionless dark matter particles, with the particle left unspecified. Only the mass profile enters, so the model says nothing about the particle's mass or identity. R5's Eddington check is what allows that reading: a spherical, isotropic, collisionless equilibrium exists for this profile inside the measured radii. Particles appear here only as the rival's language. The framework this paper imitates has none.

It passes everything by construction. Its rms to the law is 0.0000 dex, its rms to the observed curves is 0.1951, κ is 0.500 at all three cosmologies, the slope with $\rho_{\rm DE}$ is +0.500, and $a(z)/a(0)$ is 1.000 for constant $\rho_{\rm DE}$, inside the band. None of that is evidence. The formula went in, so the formula comes out.

The bill has three items. First, a particle halo has no term through which it could know $\rho_{\rm DE}$, so the tie needs a new coupling that is not one of ΛCDM's knobs. Second, no formation path is supplied, and the origin problem from R5 stands. Third, implied $\log M_{200}$ sits a median +0.11 dex from abundance matching (16th to 84th percentile −0.23 to +0.27), but −0.80 at $\log M_b$ 11 to 13. Baryon fractions are not a problem: M_b/M_{200} stays below the cosmic 0.156 in all 171.

8 Lensing

Added after version 1.0 as Amendments A4 to A6, each frozen before its numbers. Data are the KiDS-1000 profiles of isolated lenses [23]: four stellar mass bins, 60 points from 35 kpc to 2.6 Mpc, full covariance, and nothing refitted. The gate is $p \geq 0.01$. Recipes carry a Tinker bias [24] two-halo term and stop at R_{200}. The plain law has neither. The recipes, tuned only on SPARC, do not lose to the law here, and a halo made more realistic by a cutoff does worse. A post-result diagnostic, with a free mass shift in each bin and so not a grade, gives the law 167.7 on 56 degrees of freedom at +0.2 to +0.3 dex, R11 228.2 and R11b 206.0. That law number is close to an earlier, separately coded result on the same file, 163.1, in the repository's lane CFG801. All rows may share a problem with lens masses or the covariance. Satellites and the isolation cut are not modelled, recipes are built at $z = 0$, and these lenses lie mostly outside the tuned range.

Two more checks. The R11 and R11b fits are converged at 0.0415 and 0.0275 from their recorded knobs, which is a local statement. Independent halo scatter of 0.11 dex in c and 0.15 dex in M_{200}

Model	χ^2, 60 points	χ^2, $R \leq 1\,\mathrm{Mpc}$, 44 points	amplitude A
Plain law, and R12 untruncated	276.1	211.4	1.367
R12 stopped at R_{200}, two-halo	457.7	290.4	1.422
R11 twelve knobs	277.9	146.5	1.179
R11b fourteen knobs	242.1	120.9	1.140
Base ΛCDM control, untuned	563.1	444.2	0.809

Table 2: All rows fail the gate. A is one fitted scale factor, a diagnostic only; 1 is no offset.

raises their rms to observed curves only to 0.2032 and 0.1999, passing a limit of 0.205 set in advance, though noise dominates that statistic.

9 The high priest's complete recipe

R13 is one global recipe tuned jointly to the law on SPARC, to KiDS and to X-COP hydrostatic masses, frozen as Amendment A10: R11b's 14 knobs, an AGN core cutoff, lens mass offset, two-halo multiplier, satellite fraction and host mass, hydrostatic bias, and cluster baryon fraction, 21 in all, plus 5 dark energy and redshift hand terms. It was tuned on half of each dataset and graded on the rest, then tuned on all. The first run had an X-COP radius error of a factor 1000 and two smaller errors. They were fixed under the same gates and the wrong output is kept. Hand terms remain

Line	R13	Result
L1 distance to the law, limits 0.02, 0.04, 0.02	0.0290, 0.0640, 0.0187	fail, E3 passes
L2 rms to observed SPARC, limit 0.205	0.1943	pass
L3 KiDS, 60 points, $p \geq 0.01$	145.2	fail
L4 X-COP mean, rms, limits 0.05, 0.12	−0.001, 0.075	pass
L5 slope with ρ_{DE}, κ, with hand terms	+0.500; 0.483, 0.49, 0.51	pass
L6 $a(z)/a(0)$, with hand terms	0.929, 0.913, 0.92	pass
L7 Cassini, smooth halo	$2.0 \times 10^{-9}\, a_0$	pass
L8 unseen: SPARC, KiDS χ^2, X-COP	0.0281, 82.1 on 30, 0.048	fail on KiDS
L9 knobs at bounds, limit 3; median $\log M_{200}$ shift, limit 0.3	7; 0.32	fail

Table 3: R13's frozen scoreboard. Verdict: not complete.

extreme: $c \propto \rho_{\mathrm{DE}}^{3.00} H_0^{-4.86}$, $M_{200} \propto H_0^{-9.54}$. Without them the slope is +0.245. The X-COP pass rests on a cluster baryon fraction at its lower bound of 0.08, while the sample's measured value in one system is 0.138, and hydrostatic bias at 0. That is a poor physical reading, so L4's pass is weak.

R14 is the escalation, frozen as Amendment A11 after R13's verdict. SPARC uses R9, 1026 per-galaxy parameters. Lensing adds 14 per-bin parameters on R13's halo. It reaches $\chi^2 = 69.6$ on 60 points, p 0.014 at 46 degrees of freedom, and on unseen radii 44.7 on 28, p 0.024. Four of the 14 are at bounds, including truncation at $3\,R_{200}$ and the host mass. Counting R13's 21 knobs and 5 hand terms gives about 1066 parameters. It is a patchwork: nothing connects R9's per-galaxy values to the lens side, and the hand terms were not rechecked on R9's halos.

So the honest reading has two parts. Getting close is easy. Getting everything at once, as one theory, was not achieved here. For comparison, the plain law on the same X-COP points has mean −0.467 and rms 0.476 dex, and on KiDS 276.1 against R13's 145.2. On these two datasets the tuned recipe fits the data better than the law. Not graded: CMB and growth on the recipe, cluster keys,

dwarfs, IC 1613, DiskMass, the Milky Way, the Bullet Cluster, wide binaries, and any formation path.

10 If the halo were a particle

The author does not hold this reading; it is the rival's language, worked out to show what a reverse-engineered particle would need.

This is an effective reading in the rival's language. The author does not hold that the dark sector is a particle, and these results don't show that one exists. A static scalar with $y^2/(a_0 - y) = g_{\rm bar}$ reproduces the law to 3.9×10^{-13} on all 3373 points, in spherical symmetry, so $y < a_0$ always. This is not new: it is the scalar function $\mu_s = \tilde{\mu}/(1 - \tilde{\mu})$ of [25] for the simple $\tilde{\mu}$ of [26], since $s = y/a_0 = \tilde{\mu}$. If instead a real particle halo supplies all of $g_{\rm law} - g_{\rm bar}$, no force acts on baryons and double counting goes away. That halo can't bind to the Sun beyond 0.076 AU, so Cassini passes, and wide binaries should be Newtonian, unlike a field reading of the law.

A boson that condenses in all of SPARC and not in clusters at 100 kpc lands between 9.2 and 13.7 eV on our thresholds. But $\log P$ at fixed $\log \rho$ scatters by 0.521 dex across galaxies, against a limit of 0.1, so no single equation of state holds these halos up, which undercuts a simple condensate. A collisionless route from an NFW start passes the mixing theorem in 139 of 171 galaxies, all failures below $\log M_b = 10$; a frozen control there failed because our expectation was wrong, and is disclosed. The halos need a median 0.516 of all supernova energy at full coupling. Tracking baryons by scattering would need σ/m of 0.63 to 40 $\mathrm{cm^2\,g^{-1}}$. A fermion needs a mass above about 56 eV for a typical galaxy and 245 eV for the most demanding. No part of this derives $\rho_{\rm DE}$ in the halo.

11 Tying the field to dark energy

Amendment A12 writes the link as one action with one scale, $\rho_{\rm DE}$:

$$L = -\rho_{\rm DE}\left[1 + \frac{\kappa^2}{4\pi}H(s)\right] - \frac{\kappa}{\sqrt{8\pi}}\frac{\phi\,\rho_b}{M_{\rm Pl}}, \qquad s = \frac{|\nabla\phi|}{\sqrt{\rho_{\rm DE}}}, \quad H'(s) = \frac{s^2}{1-s}. \tag{2}$$

It gives $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$, with coupling $\beta = 0.0997$ and $c = 0.0199$. It reproduces the law on 3373 points to 0.00004 dex. The field's gradient energy is capped at $\rho_{\rm DE}$, and the field's own energy stays below it, $0.0579\,\rho_{\rm DE}$ at SPARC's maximum and 0.328 at 1 AU.

This is a posit, not a derivation. κ stays fitted. An earlier lane in the repository, UW1, found that this function leaves the vacuum constant free and turns complex for timelike gradients, so the action has no cosmology yet. Baryons source ϕ and so feel it, which conflicts with the particle reading unless their coupling is about 0.2 of the particle's, an order-of-magnitude figure. The field is also not the mass: a boson at $m = \rho_{\rm DE}^{1/4}$ with a potential of that height misses the dark matter density by about 49 decades.

Prior art. The capped function H was first written by the author's own programme on 22 October 2026, as the free function of an AeST embedding (lane C1), and is part of the v4 theory in PAPER59 (DOI 10.5281/zenodo.23585319). The v3 form there, $J = \Lambda - 2a_0^2[y - \ln(1+y)]$, already carries Λ and a_0 in one function. Version 1.3 of this note omitted that citation. Outside the programme, a limited literature search found no exact match for this action with $\kappa = \frac{1}{2}$, which is not proof that none exists. Close neighbours: Milgrom's $2\pi a_0 \approx c^2\sqrt{\Lambda/3}$ [14], equivalent to $\kappa \approx 0.46$; the scale $g_\Lambda = c^2\sqrt{\Lambda} = 8.4\,a_0$, where a gravitational field's energy density reaches the dark energy density [27], which is the same idea as the cap here, with β absorbing the numerical factor;

a Skordis–Złośnik extension with $J \propto Y^{3/2}/\sqrt{V_{\rm de}}$, which ties the MOND scale to the dark energy potential [28]; and unified dark sector models [29, 30]. The law itself is the simple interpolating function.

12 Can ΛCDM supply the tie itself?

Amendment A13 asks whether any acceleration scale already in ΛCDM gives $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ with no hand term. Gates frozen first: $\kappa_{\rm eff}$ within 0.05 dex of $\frac{1}{2}$, slope 0.50 ± 0.05 with $\rho_{\rm DE}$ over three cosmologies, inside the law's redshift band, one constant over $\log M$ 10 to 15, and acting on galaxy dynamics. No route passes. Milgrom's coincidence $c^2\sqrt{\Lambda/3}/2\pi$ matches the number, $\kappa_{\rm eff} = 0.461$, slope exactly +0.500, flat in z, but Λ's pull at 10 kpc is $1.1 \times 10^{-5}\, a_0$, so it acts on nothing in a galaxy; that outcome was written down before the run. $cH/2\pi$ gives $\kappa_{\rm eff} = 0.551$ but rises 3.432 times by $z = 2.33$. The NFW central acceleration $2\pi G\rho_s r_s$ is $0.830\, a_0$ at $10^{12}\, M_\odot$, but spans −0.46 to +0.48 dex over mass and scales with cosmology at slopes +1.381 and +3.561. The Tully–Fisher scale at the cosmic baryon fraction is $0.202\, a_0$ and spreads 0.505 dex. So a rival can quote Λ for the value, but no ΛCDM process here carries it into halos.

Amendment A14 tries the next step a rival would take, ΛCDM plus one named ingredient, on the same gates. None passes. Feedback as encoded in the stellar-to-halo mass relation, $V_{\rm max}^4/GM_*$ with gas left out, gives $1.715\, a_0$ at $10^{12}\, M_\odot$ but spreads 0.830 dex over mass and rises 3.200 times by $z = 2.33$. A halo edge set by Λ, at the zero-gravity radius, is flat in z but only $0.001\, a_0$ and grows as $M^{1/3}$. Coupled quintessence [1] rescales G for dark matter and carries no acceleration scale. A posited product of halo gravity and Λ's pull at the scale radius gives $0.005\, a_0$, and no ΛCDM term makes it. Knobs that carry $\rho_{\rm DE}$ remain hand terms by definition.

Correction and the strongest swoop (Amendment A15). Version 1.6 understated the halo route: A13 judged the NFW central acceleration over $\log M$ 10 to 15, wider than disc hosts. Over 11 to 12.5 it spreads only 0.089 dex, the argument of Navarro et al. [2]; this was found in an unfrozen check, then graded. Per SPARC galaxy through the stellar-to-halo relation it gives a median $0.612\, a_0$, spread 0.156 dex, $\kappa_{\rm eff} = 0.306$, and rises 1.493 times by $z = 2.33$, so it does not supply the tie. But with one global factor, 1.404, the law with each galaxy's own halo scale sits 0.0586 dex from the law and 0.1941 dex from observed curves, against the law's 0.1951. One knob on ΛCDM halos ties the law on today's rotation curves; what it lacks is the dark energy tie and a flat $a_0(z)$. On observable gates alone, since one universe has one $\rho_{\rm DE}$, Milgrom's coincidence cannot be told from the tie; only a mechanism separates them. A self-interacting core with one cross-section for all galaxies [12] fails, 0.121 to 0.126 dex from the law. Hydrodynamic simulations, fuzzy dark matter and interacting vacuum are not graded here.

13 The other dark matter models

Amendments A16 to A20 ask whether any other ΛCDM family, or any extra knob block, reaches the law exactly: rms to the law at most 0.02 dex, 95th percentile per galaxy at most 0.04, worst bin at most 0.02, and the same on unseen galaxies at most 0.025. None does, and none ties a to $\rho_{\rm DE}$.

A16 adds four knob blocks to the 14-knob recipe and keeps what helps. The best set, 15 knobs, reaches 0.0263 dex; with widened bounds 0.0252, at a median halo mass 0.64 dex above the stellar-to-halo relation. On unseen galaxies it gives 0.0267, and the per-galaxy 95th percentile, 0.053, blocks it. This is the closest single recipe so far.

A17, wave dark matter, a soliton on an NFW halo [3]: the boson mass alone gives 0.1494 dex, six knobs 0.0921. The fits improve by switching the soliton off, present in 0 of 171 galaxies at six knobs, and the soliton's acceleration scale grows with halo mass.

A18 uses the hydro-calibrated halo responses as published: Di Cintio et al. [4], coreNFW [20], and the FIRE-2 form of Lazar et al. [5]. They sit 0.116 to 0.177 dex from the law; with three knobs each, 0.073 to 0.084, wanting lighter halos. These are formulas fitted to simulations, not simulation outputs.

A19, self-interacting dark matter with a velocity-dependent cross-section, gives 0.122 dex; with three halo knobs added, 0.0797. Fitting M_{200} and c per galaxy at $\sigma/m = 0.5\,\mathrm{cm}^2/\mathrm{g}$, in the style of Ren et al. [6], gives 0.0487 with 343 parameters, 77 at a bound, and 0.110 on unseen galaxies at prior centre. Warm and mixed warm plus self-interacting dark matter give 0.116 to 0.120. A coding error in the first control was fixed; the wrong output is kept.

A20, interacting vacuum, $Q = \xi H \rho$ [7], is the only route that puts ρ_{DE} into the halos. Near $\xi = -0.35$ it gives κ_{eff} 0.456 to 0.550, but the halo scale drifts with redshift, spreads 0.152 dex over galaxies, and put into the law it sits 0.0577 dex away. (Versions 1.8 to 2.0 printed $\xi = -0.55$, 0.153 and 0.0576: the growth equation used for this row had the wrong damping term, found by a later lane and fixed; the verdict is unchanged, and the wrong output is kept.) A20 also asks whether the law is only a reading of measurement error on ordinary halos. Mocks with SPARC's errors match the total scatter, a weak test since the errors drive it, but a fitted a_0 comes out +0.228 and +0.415 dex high, without and with contraction.

So the closest a rival gets is 0.0252 dex with 15 knobs and stretched halos, and the only tie to ρ_{DE} comes from a coupling that drifts in z. Coefficients for the wave and warm dark matter rows are from memory and were not checked against the papers. Real hydrodynamic simulation outputs are not graded here.

14 Simulations and the near-MOND dark sectors

Amendments A21 to A24 grade four further routes on the same gates. None gives the tie natively.

A21 asks whether the law simply emerges in ΛCDM hydrodynamic simulations, using published results. No measured simulation scale reaches the law: the best is Magneticum, 0.0290 dex, then MassiveBlack-II faint galaxies 0.0306 and ZOMG 0.0359; EAGLE's raw fits sit 0.17 to 0.19 away. Only round order-of-magnitude figures of $10^{-10}\,\mathrm{m\,s^{-2}}$ pass. The EAGLE runs with varied Λ report no acceleration relation, so the slope with ρ_{DE} cannot be tested from any simulation on record, and where redshift numbers exist the scale rises into the past, 1.305 to 1.428 times in ZOMG. These values are read from the papers' summaries and are provisional.

A22, superfluid dark matter [8]: the deep phonon force added to baryons, $g_b + \sqrt{a\,g_b}$, sits 0.0392 dex from the law, 0.0378 on unseen galaxies; at the published fiducial values with the condensate's gravity 0.0810; with four knobs 0.0389; with 346 parameters 0.0370. Setting its energy scale by $\Lambda^4 = \rho_{\mathrm{DE}}$ gives κ_{eff} 5.014; reaching 0.5 needs its coupling set by hand, so κ moves into that coupling, and nothing in the model pins the scale to ρ_{DE}. The matching rule was changed after a first test matched none of 171 galaxies; this is disclosed.

A23, dipolar dark matter [30]: the paper's cubic function gives the same $g_b + \sqrt{a\,g_b}$, 0.0392. Its free function can, however, take the law's shape exactly: the law is one such choice, and a fourth-order expansion with three knobs reaches 0.0036, 0.0033 on unseen galaxies. That is a fit of shape in a reformulation of MOND, not a ΛCDM mechanism. Its link to Λ is stated as an order of magnitude and imposed; at the quoted coupling κ_{eff} is 0.576, and as written the slope with ρ_{DE} is 0.

A24, emergent gravity [9]: in no reading does it reach the law, best 0.0392 with one knob, 0.0378 on unseen galaxies. With today's H_0 it gives $\kappa_{\rm eff}$ 0.583 and rises 3.510 times by $z = 2.33$. With the de Sitter rate $\sqrt{\Lambda/3}$ it gives $\kappa_{\rm eff}$ 0.482, slope 0.5 and no drift, but the effective scale spreads 0.115 dex between galaxies, and the choice of rate is imposed: it is the Λ coincidence of Amendment A13 with 1/6 in place of $1/2\pi$.

Three different theories reduce in their plain forms to $g_b + \sqrt{a\, g_b}$, which misses the law by the same 0.0392 dex because its transition has the wrong shape.

15 The law and the tie together

Amendments A25 to A28 aim at the law and its tie to dark energy at once. None reaches both.

A25 stacks 28 ΛCDM ingredients into one global recipe: the 14 feedback knobs, the A16 blocks, coreNFW and DC14 responses, a self-interacting core, adiabatic contraction, separate settings for gas-rich and bulge galaxies, and the slope and scatter of the stellar-to-halo relation. At no knob count is it exact. The curve of distance against knob count flattens near 0.022: with 28 knobs 0.0222, per-galaxy 95th percentile 0.0546. With widened bounds it reaches 0.0204, but 0.0370 on unseen galaxies, at a plausibility cost of 1.290 range widths outside the literature. Its $\kappa_{\rm eff}$ of 0.497 is inherited from the target; its scale runs with cosmology and rises about three times by $z = 2.33$. In 24 % of the self-interacting core solves the core did not converge; those points were regraded with a direct solver.

A26 puts the frozen 14-knob recipe into other universes. Scaling $\rho_{\rm DE}$, its scale moves with slope +0.1874 to +1.4526 depending on what is held fixed, against 0.5, and by $z = 2.33$ it is 3.0235 to 4.5549 times today's. Added dependences can fake both gates only with a term written in $\rho_{\rm DE}$; the expansion rate alone reaches a slope of +0.4302, and the faked recipes lose the law's shape at high redshift.

A27 enumerates accelerations built from ΛCDM's constants: 199584 expressions. 104 of them, 63 distinct values, pass the number, slope and redshift gates, against 62.1 expected by chance with the target slid off its value. Most are the Λ coincidence dressed with other ratios; none reaches galaxy radii.

A28 fits each galaxy's observed curve with mass-to-light ratio, distance and inclination free within published priors. Here the rival claim holds: cored halos fit better than the tied law even after the parameter penalty, BIC 17804.6 against 19744.2, and beat it in 154 of 171 galaxies. Their fitted galaxies, however, give a relation +0.056 to +0.103 dex off the tied a_0 with 0.1254 to 0.1368 dex scatter, so they reproduce the data without reproducing the law. A better fit is not a derivation, and it is not proof for either side. The prior widths are recalled, not checked.

So the swoop that works is fitting the curves galaxy by galaxy; the one that fails is deriving the law and its tie.

16 Evolving dark energy and the rival literature

Amendments A29 to A40 add one gate: where dark energy evolves, the scale must follow $\sqrt{\rho_{\rm DE}(z)}$ within 0.05 dex at $z = 1$, 1.5 and 2.33, as the tie predicts. No route reaches both the law and the tie.

In evolving backgrounds (A29) ΛCDM halo scales follow the expansion rate and the matter density, not dark energy: pooled across backgrounds their slope with $\rho_{\rm DE}$ runs from −0.1241 to +0.0256, and none of 315 w_0–w_a backgrounds passes with the 14-knob recipe frozen. Only the Λ coincidence, generalised to $\rho_{\rm DE}(z)$, tracks, by construction, and it acts on nothing at galaxy radii.

Tuning feedback knobs and dark-energy parameters together (A32) reaches tracking only through fitted powers of $H(z)$ with no named process (stellar-to-halo relation $^{+3.017}$, feedback $^{+4.245}$), which distort the curves at high redshift and fail on unseen galaxies.

Screened fields (A30): a three-knob k-mouflage fit with no halo reaches the law's shape, 0.0086 dex and 0.0091 on unseen galaxies, because its switch lies below almost every point and it acts as a power law; it adds $61.6\,a_0$ at Saturn and its scale does not depend on $\rho_{\rm DE}$. Chameleon, symmetron and coupled quintessence fail. Time-varying energy exchange, running vacuum and Chaplygin gas (A31) give no setting that passes κ and tracking together.

The ΛCDM papers that explain MOND phenomenology (A34): the cooling scale of Kaplinghat and Turner [10], put inside the law's formula and multiplied by one factor, 0.398, set by hand, matches today's curves exactly, 0.0152 dex and 0.0156 on unseen galaxies. The shape is borrowed from the law and the scale is tuned; it follows the matter density, not $\rho_{\rm DE}$, and rises 1.4231 times by $z = 2.33$. Disc-formation and halo-response papers miss by 0.12 to 0.22 dex.

Every place $\rho_{\rm DE}$ enters ΛCDM structure (A35) gives no tie; the native collapse-epoch route gives $\kappa_{\rm eff}$ 0.4358. The tie contains c, but halo accelerations do not depend on c: at $10^{12}\,M_\odot$ the ratio $c/V_{200} = 2102$ is bridged by overdensity and concentration factors that carry no c, so the halo scale's closeness to a_0 is a coincidence. The logotropic fluid [11] (A36) carries the slope 0.5 natively, but its halo pull caps at $0.298\,a_0$ where the law needs $0.618\,a_0$, and its scale stays flat as dark energy evolves. De Sitter and quantised inertia (A37) share one shape, 0.0189 dex with a free constant but failing the worst bin at 0.0273, with $\kappa_{\rm eff}$ 0.6506. Dark matter tied to baryons (A38) reaches 0.0715 for dissipative dark matter and 0.1864 for dark-matter–baryon thermalisation as published. Clustering dark energy and dark-energy drag (A39) put at most $10^{-1}\,a_0$ into halos. Correlated halo scatter and microphysical constants (A40) leave a_0 too high, best 0.1539 dex, with $\kappa_{\rm eff}$ 0.796.

A growth-equation error in two earlier lanes was found by A32 and fixed; it moved one coupling value and no verdict. Many literature values here come from summaries of the papers and are provisional.

17 The remaining dark matter families

Amendments A41 to A44 grade every dark matter family left open on the folder's coverage map. None reaches the law, and none ties a to $\rho_{\rm DE}$: their scales carry no $\rho_{\rm DE}$, rise by factors of 1.4 to 5 by $z = 2.33$, and fail the tracking gate in every evolving background.

Self-interacting dark matter on its collapsed branch sits 0.1790 dex from the law; with each galaxy free to choose its branch and halo, 514 parameters, 0.0472, but 0.110 on unseen galaxies. Condensate dark matter with a fixed core radius gives 0.1289 and has no core solution in 83 of 171 galaxies. Mixed cold and fuzzy dark matter gives 0.1174, cold and warm 0.1157 (A41).

Decaying dark matter, primordial black holes, dark matter coupled to dark radiation and a dark-matter-only Yukawa force (A42) reach at best 0.0717, with a repulsive force at its bound; each fitted particle parameter lies beyond published limits, for example black holes of $4 \times 10^5\,M_\odot$ that would also overheat the discs. Dark mass that traces the gas (A43) gives 0.4066 as published and 0.2084 with a stellar term, because bright discs hold too little gas; dark-matter–baryon scattering gives 0.1299 at a cross-section 104 times the CMB limit.

Halo flattening, tidal truncation, substructure and environment (A44) move the best 28-knob recipe from 0.0222 to 0.0221 dex, so the gap to an exact match stays. Many limits and coefficients in these lanes come from summaries or memory and are provisional.

18 Modified gravity and measurement systematics

Amendments A45 to A48 grade the modified-gravity routes a rival could take, and the reading that the law is a product of measurement systematics on ordinary halos. None gives the law with a native tie to $\rho_{\rm DE}$.

Vainshtein-screened DGP and Galileon gravity (A45) leave galaxies Newtonian, 0.4956 dex from the law, or 0.1424 with a ΛCDM halo; its deep force has a square-root form but a scale that grows with radius. Beyond-Horndeski theories have no acceleration scale, best 0.0990 with a halo. Mashhoon's non-local gravity misses by 0.3172 as published. In relativistic MOND and bimetric MOND (A46) a_0 is a constant of the Lagrangian; TeVeS's standard function is $g_b + \sqrt{a_0 g_b}$, 0.1006 at its usual a_0, and where the interpolation is free it can be chosen to be the law, a choice rather than a derivation. MOG misses by 0.3769 and conformal gravity by 0.1360 as published.

Non-circular motions, bars, pressure support and disc thickness (A47), applied to ΛCDM mocks at literature amplitudes, move their fitted scale from +0.240 to +0.101 dex above the tied a_0 and loosen the relation. Beam smearing, selection, mass-to-light trends and gas uncertainties (A48) reach at best 0.1222, 0.1316 and 0.1416 singly; every mock keeps an excess of +0.13 to +0.16 dex in the lowest-acceleration bin. A selection weight toward weak halos reaches the scale, $\kappa_{\rm eff}$ 0.518, but not the shape. The same corrections, applied to SPARC, raise the law's own fitted a_0 by up to +0.101 dex, and allowed mass-to-light and gas changes move it by −0.174 to +0.157 dex: the fitted $\kappa = \frac{1}{2}$ carries that systematic uncertainty. Amplitudes and ranges are provisional.

19 Reading the menu

The point of the exercise is how little it takes. A dozen feedback knobs get the level, nearly the shape, and a tie with the law on observed curves. So a future ΛCDM paper showing a tight relation at this scale has not, on that alone, derived anything. A paper claiming to derive this law from ΛCDM should show these five things: the prior on M_{200} and c per galaxy, the fit against surface-density diversity and not only mass, the slope with $\rho_{\rm DE}$ with baryons evolved, κ across cosmologies, and a formation path to the required halo. Failing them doesn't refute ΛCDM, since real simulations could do things this model can't. Passing them would be a real result, and the law would then have to be read as an output of the halo side.

20 Limits

This is a semi-analytic model, not a simulation. Baryons are spherically approximated in the halo response. Mass-to-light ratios are fixed. Priors and knob ranges are our judgement. The feedback mapping is ours and has not been tested against any simulation, so whether real subgrid physics can reach these knob values is open. Masses M_* use one Υ and one SHMR. Among clusters only X-COP is graded. A coding error in the cosmology rows was found and fixed during this work; the wrong output is kept in the repository. Three coding errors in the lensing two-halo code were fixed before its first full table. All numbers are AI-run and have not been reproduced independently.

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